Majid Abbasi Sisara
Ph.D. in BKN
University of Southern California
Personal Website

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Google Scholar

PROFESSIONAL SUMMARY

Experienced Computational Neuroscientist with a strong background in biomedical engineering, machine learning, and digital signal processing. I am skilled in the analysis and interpretation of complex brain data from diverse sources, including LFP, ECoG, EEG, fMRI, and DTI, with a particular focus on brain mapping and neuroimaging. Strong expertise in developing and deploying machine learning models, particularly for brain-computer interaction and cognitive neuroscience applications. Throughout my career, I have contributed to interdisciplinary research projects, leveraging big data analytics to improve outcomes in brain health, cognitive function, and mental well-being. Committed to advancing the integration of AI in healthcare and utilizing brain mapping techniques to enhance our understanding of neurological conditions.

Education

•Ph.D. in Biokinesiology *2022-2027*
University of Southern California (USC) GPA: 3.94/4

•M.Sc. in Biomedical Engineering *2013-2015*
Sharif University of Technology (SUT) GPA: 17/20

•B.Sc. in Electrical Engineering *2009-2012*
Tabriz University of Technology (TUT) GPA: 18.06/20

Publications

•Conference Abstracts

- **Abbasi Sisara, M.**, Cohen, S., Azadjou, H., Duncan, D., Liu, C., & Valero-Cuevas, F. J. (2025). The bilateral brain networks for dexterous manipulation are severely compromised by TBI: An identical-twins study. *Society for Neuroscience Annual Meeting*, San Diego, CA.
- **Abbasi Sisara, M.**, Salaway, M., Hajian Berenjestanaki, M., & Valero-Cuevas, F. J. (2025). The Impact of Different Functional Connectivity Measures on the Classification of Harmful Brain Activity Using Various Graph Neural Networks. *Society for the Neural Control of Movement*, Panama City, Panama.
- Azadjou, H., Philip, R., **Abbasi Sisara, M.**, Seki, K., & Valero-Cuevas, F. J. (2025). Neural dynamics during reach-to-precision-grasp behavior emerge from nonlinear recurrent cortical activity. *Society for Neuroscience Annual Meeting*, San Diego, CA.
- **Abbasi Sisara, M.**, Azadjou, H., Kikuta, S., Kubota, S., Seki, K., & Valero-Cuevas, F. J. (2025). The brainstem modulates sensory transmission to S1 in task-dependent ways. *Society for the Neural Control of Movement*, Panama City, Panama.

•Journal Papers

- Jamali, A., **Abbasi Sisara, M.**, Nasab, E. K., Van Dam, D., & Amiri, M. (2026). G-ALPS: An index for evaluating cognitive decline and aging using diffusion tensor imaging. *Alzheimer's & Dementia*, 22(3), e71118.
- Esmaeil-Zadeh, M., Fattahi, M., Rasouli, N., Soltanian-Zadeh, H., **Abbasi Sisara, M.**, Rajab, E., ... & Malakouti, S. K. (2025). fMRI features in recent suicide attempters performing the future imagination task. *Psychiatry Research: Neuroimaging*, 112049.
- Mohammadian, F., Zare Sadeghi, A., Noroozian, M., Malekian, V., **Abbasi Sisara, M.**, Hashemi, H., ... & Saligheh Rad, H. (2023). Quantitative assessment of resting-state functional connectivity MRI to differentiate amnestic mild cognitive impairment, late-onset Alzheimer's disease from normal subjects. *Journal of Magnetic Resonance Imaging*, 57(6), 1702-1712.
- Cuzzocrea, A., **Abbasi Sisara, M.**, Leung, C. K., Wen, Y., & Jiang, F. (2022, July). Effectively and Efficiently Supporting Visual Big Data Analytics over Big Sequential Data: An Innovative Data Science Approach. *In International Conference on Computational Science and Its Applications* (pp. 113-125). Cham: Springer International Publishing.
- Cuzzocrea, A., **Abbasi Sisara, M.**, & Gallo, C. (2022, June). Experimental analysis and verification of a multi-modal-biometrics identity verification framework based on the Dempster-Shafer theory. *In Computational Intelligence in Security for Information Systems Conference* (pp. 118-129). Cham: Springer International Publishing.
- Cuzzocrea, A., **Abbasi Sisara, M.**, Lenac, K., & Mumolo, E. (2022). Supporting Big Moving Objects Tracking and Analysis: An Innovative Scan-Matching Algorithm. In *SEBD* (pp. 404-414).

– Taheri, M. A., Torabi, S., Nabavi, N., Modarresi-Asem, F., **Abbasi Sisara, M.**, Rezaei, A., ... & Semsarha, F. (2021). Task-fMRI Group and Functional Connectivity Analysis of the Brain During Faradarmani Consciousness Field Connection.

•Submitted Papers

– **Abbasi Sisara, M.**, Kubota, S., Azadjou, H., Kikuta, S., Seki, K., Valero-Cuevas, F. (2026). Skin and muscle sensory signals in primates are transmitted and modulated in distinct, active, and task-dependent ways. *Science Advances(AAAS)*.
– **Abbasi Sisara, M.**, Azadjou, H., Cohen, S., Y. Liu, C., Diaz-Arrastia, R., Duncan, D., Valero-Cuevas, F. (2026). Dexterity-Dependent Bilateral Brain Networks in Monozygotic Twins Discordant for Traumatic Brain Injury: An fMRI Case Report. *Neurology Open Access*.
– **Abbasi Sisara, M.**, Y. Liu, C., Diaz-Arrastia, R., Valero-Cuevas, F. (2026). Structure versus Dynamics: Regional Diffusion Metrics Detect Traumatic Brain Injury While DTI-ALPS Tracks Recovery. *Neurology*.
– **Abbasi Sisara, M.**, Salaway, M., Valero-Cuevas, F. (2026). The Impact of Different Functional Connectivity Measures on Classification of Harmful Brain Activity Using Various Graph Neural Networks. *IEEE*.
– Bertucco, M., **Abbasi Sisara, M.**, Benamati, A., Trinchi, M., Cesari1, P., Valero-Cuevas, F. (2026). Dexterous skill acquisition under neuromuscular fatigability leaves a lasting frequency-specific signature in the common neural drive to hand muscles. *Frontiers In Systems Neuroscience*.

•Pre-Print Papers

– Razmara, P., Medani, T., **Abbasi Sisara, M.**, Joshi, A. A., Chen, R., Jeong, W., ... & Leahy, R. M. (2026). Feasibility of simultaneous EEG-fMRI at 0.55 T: Recording, Denoising, and Functional Mapping. *arXiv* preprint arXiv:2602.13489.
– Razmara, P., Medani, T., Joshi, A. A., **Abbasi Sisara, M.**, Tian, Y., Cui, S. X., ... & Leahy, R. M. (2025). A feasibility study of task-based fmri at 0.55 t. *arXiv* preprint arXiv:2505.20568.
– **Abbasi Sisara, M.**, Hosseini, S. H. K., Mahjoub, H., Kordnaeij, A., & Esmaeilinasab, M. (2024). Behavioral and neural correlates of temperament traits: insights from temperament and character inventory and fMRI-Based choice tasks. *bioRxiv*, 2024-11.

Experiences

•Research Assistant *Los Angeles, CA*
Aug 2022 – May 2027 University of Southern California

– **Hand Manipulation:** A multimodal study using EEG, EMG, and fMRI to understand coherence and correlation between brain regions (brain networks) and hand muscles during performing a dexterity task by hand. The participants are traumatic brain injury (TBI) and healthy controls.
– **Locomotion and Leg Dexterity:** A multimodal study using EEG and EMG to understand coherence and correlation between brain regions (brain network) and leg muscles during performing a leg dexterity task. The participants are stroke patients and healthy controls.
– **Multimodal Data Analysis:** Conducted data analysis on Diffusion tensor imaging (DTI), resting state fMRI (rs-fMRI), task-based fMRI, dexterity-dependent task-based EEG and EMG, and resting state EEG of traumatic brain injury (TBI) patients.
– Analyzed complex motor control mechanisms in 10 participants with drop foot using dimensionality reduction, identifying critical control deficiencies and suggesting targeted interventions
– **Machine Learning:** Developed and applied 4 deep learning models (Variational RNN autoencoders, transformers, GNNs, and MLP mappers) to classify harmful brain activity
– **Mentoring Interns:** Mentored 7 graduate interns, 4 undergrads from various backgrounds (EE, CS, and BME)
– **Writing Proposal (R01 and R21):** Contributed in writing 4 research funding proposals for NIH and NSF

•Computer Scientist *Trieste and Rende, Italy*
Jan 2022 – Aug 2022 University of Trieste and University of Calabria

– Big Data Engineering and Analytics Laboratory (iDEA Lab) | Advisor: Prof. Alfredo Cuzzocrea
• Big Data Analysis • Multimodal-Biometrics • Moving Objects Tracking

•Image Processing Specialist (Data Scientist) *Tehran, Iran*
Sep 2019 – Sep 2022 National Brain Mapping Lab (NBML)

– Brain mapping experiment design | Tools: PsychoPy and Psychtoolbox
– Presurgical Planning (task-based fMRI, rs-fMRI, and DTI)
– fMRI data analysis | Tools: SPM and FSL

– Resting-state fMRI data analysis | Tools: CONN and FSL
– DTI image analysis | Tools: ExploreDTI, Dai-Stodio, FSL
– EEG-fMRI data analysis
– Eye tracking (experiment and analysis)
– Sleep Analysis (EEG)

•Data Scientist *Tehran, Iran*

Dec 2016 – Sep 2019 Sharif University of Technology

– Electronics Research Institute | Advisor: Prof. Siavash Bayat
• Ultrasound Signal Processing • Algorithm Design • Smart-City

•Data Scientist - Research Assistant (RA) *Tehran, Iran*

June 2019 – Sep 2019 Institute for Research in Fundamental Science (IPM)

– School of Cognitive Science; Training monkeys, recording data, and spike sorting by a 10*10 Utah array in PFC | Advisor: Dr. Ali Ghazizadeh

•Research Assistant (RA) *Tehran, Iran*

Sep 2018 – April 2019 Sharif University of Technology

– ChabokLab (Ultrasonic Lab), Department of Physics | Advisor: Dr. Hamid Chabok

•Teaching Assistant *Los Angeles, CA*

Aug 2022 – Dec 2026 University of Southern California

– Musculoskeletal Anatomy Lab | Instructor: Dr. George Salem, Dr. Susan Sigward, Dr. Kate Havens
– Special Topics in Therapeutic Exercise | Instructor: Dr. Kenneth Kim, Dr. Liz Poppert, Dr. Rob Landel

•Instructor *Tehran, Iran*

Feb 2017 – June 2019 Sharif University of Technology, Department of Electrical Engineering

– Linear Control Lab

•Instructor *Tehran, Iran*

Spring 2017 University of Applied Science and Technology, Department of Biomedical Engineering

– Bioinstrumentation (3 units)
– Bioinstrumentation Lab (1 unit)
– Troubleshooting of Medical Equipment (3 units)
– Troubleshooting of Medical Equipment Lab (1 unit)

•Teaching Assistant *Tehran, Iran*

Fall 2014 Sharif University of Technology, Department of Electrical Engineering

– Physiological Systems Modeling | Instructor: Dr. Mehran Jahed

•Teaching Assistant *Tabriz, Iran*

Feb 2010 – June 2013 Sahand University of Technology, Department of Electrical Engineering

– Microprocessor | Instructor: Dr. Akbar Ghaffarpour
– Pulse Technique | Instructor: Dr. Hasan Faraji
– Electronic circuits | Instructor: Dr. Esmaeil Najafiaghdam
– Signals & Systems | Instructor: Dr. Mohamad Hosein Sedaghi
– Electrical circuits | Instructor: Dr. Reza Mahboobi
– Electrical circuits Lab | Instructor: Dr. Reza Mahboobi
– Electronic circuits Lab | Instructor: Dr. Reza Mahboobi
– Signals & Systems | Instructor: Dr. Mousa Shamsi
– C Programming Language | Instructor: Dr. Mina Salim

•Peer Reviews *Los Angeles, CA*

Aug 2023 – Present University of Southern California

– Science Advances (AAAS) (2025)
  ∗ Linear Acceleration Is a Primary Risk Factor for Concussion and a Target for Prevention
– Transactions on Neural Systems and Rehabilitation Engineering (2025)
  ∗ Hand Function Assessment via PCA to Capture High Dimensional Kinematic Signals

– PLOS ONE (2024)
  * Transfer Learning and Reweight Examples for Robust Classification of Small-sample Dataset
  * In vivo mapping of sodium homeostasis disturbances in individual ALS patients: a brain 23Na MRI study
  * Development of clinical decision support for patients older than 65 years with fallrelated TBI using artificial intelligence modeling
  * A 10 Years Update of Effects of Exercise on Depression Disorders - in otherwise healthy adults: A Systematic Review of Meta-Analyses and Neurobiological Mechanisms
  * Fortifying Convolutional Neural Networks Against Adversarial Attacks: A Comprehensive Evaluation and Enhancement of Defense Mechanisms
  * Leveraging Transformers and Explainable AI for Alzheimer's Disease Interpretability
  * RRFNet: A free-anchor brain tumor detection and classification network based on reparameterization technology
  * Prediction of heart disease based on stacking algorithm
  * From data to diagnosis: an innovative approach to epilepsy prediction with CGTNet incorporating spatio-temporal features
– Springer Nature (SNAPP) (2024)
  * Functional Connectivity Changes in Human Brain Networks: from 2 Hz Rhythmic Muscle Contraction to the Hand: A pilot study
– International Journal of Robust and Nonlinear Control (2023)
  * Modeling of Isometric Muscle Properties via Controllable Nonlinear Spring, and Hybrid Model of Proprioceptive Receptors

•Volunteering *Los Angeles, CA*

Jan 2023 – Mar 2023 NSF DARE—Transforming modeling in neurorehabilitation

– Led the strategic planning and coordination of logistics for over 120 posters and presentations, ensuring smooth execution and effective event management

Professional Certificates

– Health, Technology, and Engineering (HTE) certificate, USC
– Health Insurance Portability and Accountability Act (HIPAA) training certificate, USC

WORKSHOPS AND WEBINARS

– "AI and Brain Computer Interfaces: Connecting the human mind to intelligent machines" Share talk with Dr. Mohsen Attaran, Talangor, Los Angeles, CA, November 2026.
– "Human Brain Mapping: a windo from brain to mind" Talk, Talangor, Los Angeles, CA, June 2026.
– "DTI based Alzheimer's disease classification" webinar, Brain Mapping Laboratory, Iran, February 2021.
– "Alzheimer's Disease stage identification using Deep Learning models" webinar, , Brain Mapping Laboratory , Iran, November 2014.
– "fMRI Experiment Design" workshop, Brain Mapping Laboratory, Tehran, Iran, January 2021.
– "Principles of Eye Tracking and Its Applications in Brain Mapping" webinar, Brain Mapping Laboratory, Iran, May 2020.
– "Introduction of Task Design in fMRI" webinar, Brain Mapping Laboratory, Iran, July 2020.

Skills

Programming: Python (Advanced), MATLAB (Expert), Verilog (Experienced), C++ (Proficient), R, SQL
ML/AI Frameworks: PyTorch, TensorFlow, Keras, OpenAI Baselines, NumPy, SciPy, Scikit-learn
Reinforcement Learning: On-policy/off-policy methods, policy gradients, actor-critic, PPO, imitation learning
Generative Modeling: Variational Autoencoders (VAEs), Transformers, Latent Dynamics Modeling
Multimodal AI: Vision, tactile (EMG), and language-conditioned sensor fusion
Data & Signal Analysis: Adaptive filtering, synchronization analysis, statistical learning from neural and biosignals
Software Engineering: FSL, SPM12, AFNI, FreeSurfer, PsychoPy, Psychtoolbox, CONN, ExploreDTI, ANT, Dsi-Studio, EEGLab
Languages: English (Fluent), Persian (Native)

Coursework

– Deep Learning
– Machine Learning for Data Science
– Artificial Intelligence and Biological Computations
– Neuromuscular and Bio-Robotic Systems
– Foundations of Systems and Computational Neuroscience
– Digital Signal Processing
– Medical Imaging Systems
– Physiological Signals and Data Analytics